

Measuring and removing the costs of parametric Gröbner bases for generated inverse kinematics

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Abstract

Inverse kinematics can be posed with the target adjoined to the coefficient field rather than substituted into the polynomial ring, so that one Gröbner basis over $\mathbb{Q}(\mathbf{p})$ answers every pose and its action matrices can be emitted as source code. In the implementation studied here that computation completed in milliseconds with two adjoined parameters and never with three, although a tool position has three coordinates. We locate the cost, remove it, and locate and remove the cost that appears behind it. Profiling attributes 86% of the time to the polynomial gcd used to keep coefficients in lowest terms; repeating the computation over $\mathbb{F}_p(\mathbf{p})$, and measuring the subresultant gcd in isolation over both fields, shows that the cost grows as the 3.8th power of the operand size whatever the field, so that only a gcd built on evaluation and interpolation can help. Brown’s dense modular algorithm, certified by exact division over $\mathbb{Z}$ and returning the cofactors that division computes, brings the exponent to 1.47 and a gcd of 455-term operands from 91 s to 4.6 ms, and the three-parameter solve of the anthropomorphic arm, which had not finished in 900 s, to about 20 s. On arms with offsets it still does not finish in an hour, because the parameter polynomials carried through the completion swell to 16 000 terms while the matrices finally produced have entries of fourteen. Reconstructing those entries from fixed-pose solves over prime fields, by rational function interpolation, never forms the intermediate expressions and solves every arm tested, three of which the usual decoupling of the base joint cannot treat, in 0.25 s to 47 s; each result is checked exactly over $\mathbb{Q}$ at rational poses and agrees entry for entry with the symbolic solve where that finishes. Finally, a Newton step on the equations as posed removes the residual tail of the emitted solvers, whose worst error falls from 6.5×10^{-8} m to 1.6×10^{-15} m at a cost of 0.6 μ s per call.

1 Introduction

The inverse kinematics of a serial manipulator with revolute joints becomes a polynomial system under the tangent half-angle substitution $t_i = \tan(q_i/2)$, and the resulting zero-dimensional ideal can be solved by the eigenvalue method of Möller and Stetter [14]. Classical treatments of the general $6R$ arm reduce the problem to a univariate resultant of degree sixteen [12, 19, 13, 10], and code generators such as IKFast [6] match an arm against a catalogue of solvable structures. The system studied here takes a different route. The target pose is adjoined to the coefficient field, so that the Gröbner basis is computed once for the manipulator and evaluated at each requested pose, and the action matrices of the quotient algebra are emitted as source code. The number of solutions is then the dimension of the quotient algebra, a certificate that no branch has been missed, and a call to the emitted solver costs about as much as a few evaluations of the forward kinematics.

The obstacle is the cost of the offline computation. Writing N for the number of actuated joints and P for the number of adjoined pose coordinates, only $P = N$ can give a finite, nonempty solution set (Section 2), and a tool position has $P \leq 3$. The implementation completed $P = N = 2$

in tens of milliseconds and had produced no result for $P = N = 3$ after 900 s, which left every three-joint arm to a decoupling that applies only when the base placement commutes with the base rotation. This paper follows the cost of the three-parameter problem through two bottlenecks and removes each.

Contributions.

1. A controlled measurement that separates the two usual explanations of a slow gcd over $\mathbb{Q}(\mathbf{p})$ (Sections 3–5). Substituting $\mathbb{F}_p$ for $\mathbb{Q}$ does not make the problem finish, and the subresultant gcd costs the same power of the operand size over both fields, so that the coefficient field governs a bounded constant and computing the same remainder sequence modulo several primes cannot help.
2. An implementation of Brown’s dense modular gcd [2] in which the certifying division is carried out over $\mathbb{Z}$ and its quotients are returned as cofactors, with a correctness argument for the leading-monomial rule it relies on (Section 6), and a measurement of its effect in isolation and on the parametric solve, with ablations (Section 9).
3. The observation that, once the gcd is removed, the cost of the parametric solve on arms with offsets lies in intermediate expression swell, and a reconstruction of the action matrices from fixed-pose solves over prime fields [11, 21, 17] that avoids it, checked exactly at rational poses (Section 7). Every three-joint arm tested, including three the decoupling cannot treat, one of them with axes in general position, is solved in under a minute.
4. A Newton refinement on the equations as posed, emitted into the generated solver, which removes the error the eigenvalue method incurs through ill-conditioned eigenvectors (Section 8).

Sections 3 to 5 give the diagnosis in the order in which it was carried out, beginning with the design of the field substitution and continuing with its results and with what they imply for the choice of gcd. Section 6 states the gcd adopted, Section 7 the reconstruction, Section 8 the refinement, and Section 9 measures all three. The implementation is part of the *varietas* library, available under the Apache License 2.0 at <https://github.com/HanielUlises/varietas>, and the appendix lists the programs that produce every figure reported.

2 Preliminaries

2.1 Notation

Let k be a field and $k[\mathbf{t}] = k[t_1, \dots, t_N]$ the polynomial ring in the joint variables. For a monomial order $\prec$ and nonzero $f \in k[\mathbf{t}]$, $\text{LM}(f)$ denotes the leading monomial of f , and for an ideal I the initial ideal is $\text{in}(I) = \langle \text{LM}(f) : f \in I \setminus \{0\} \rangle$. A finite $G \subset I$ is a Gröbner basis when $\text{in}(I) = \langle \text{LM}(g) : g \in G \rangle$. The quotient $A = k[\mathbf{t}]/I$ is a k -vector space with basis the monomials outside $\text{in}(I)$, called the standard monomials, and I is zero-dimensional exactly when that basis is finite, in which case $\dim_k A$ bounds the number of points of $V(I)$ over $\bar{k}$ and equals it when I is radical. We refer to [5] throughout for these facts and to [3] for the completion algorithm.

Saturation is written $I : h^\infty$. The half-angle substitution introduces denominators $1 + t_i^2$, and clearing them attaches to $V(I)$ the loci $t_i = \pm\sqrt{-1}$, which correspond to no configuration of the manipulator. Saturating by $h = \prod_i (1 + t_i^2)$ removes them, so that $\dim_k A$ counts configurations alone.

2.2 The parametric formulation

Definition 1 (Parametric position problem). Let $\Phi : \mathbb{A}^N \rightarrow \mathbb{A}^3$ be the rational forward map of a manipulator with N actuated joints, written over the coefficient field k , with numerator components Φ_i and common denominator D . Let $C \subseteq \{1, 2, 3\}$ with $|C| = P$ index the constrained tool coordinates, and adjoin indeterminates $\mathbf{p} = (p_1, \dots, p_P)$ to k . The parametric position ideal is

$$I = \left\langle \Phi_{c_j}(\mathbf{t}) - D(\mathbf{t})p_j : j = 1, \dots, P \right\rangle : h^\infty \subseteq k(\mathbf{p})[\mathbf{t}].$$

Coefficients are elements of $k(\mathbf{p})$, represented as a quotient of two polynomials in $\mathbf{p}$ over k and reduced to lowest terms after every arithmetic operation. Reduction requires a polynomial gcd in $k[\mathbf{p}]$, computed at the outset of this work by a subresultant remainder sequence in the sense of Collins [4], and by the construction of Section 6 at its end. Without it the representations grow without bound under the repeated additions and multiplications of the completion algorithm, so the reduction is a requirement of the formulation rather than an economy.

2.3 The admissible number of parameters

The two results below restrict attention to $P = N$ and identify the three-parameter case as the smallest spatial instance.

Proposition 2. *Let I be as in Definition 1 with $P < N$. Then every irreducible component of $V(I)$ has dimension at least $N - P$, and I fails to be zero-dimensional unless $V(I) = \emptyset$.*

Proof. The ideal before saturation is generated by P elements, so by Krull's height theorem [7, Ch. 10] every minimal prime over it has height at most P , whence every component of its zero set has dimension at least $N - P$. Saturation deletes components and leaves the dimension of those that survive unchanged, so the bound persists. The degenerate case $V(I) = \emptyset$ arises when saturation deletes every component. $\square$

Proposition 3. *Let I be as in Definition 1 with $\mathbf{p}$ algebraically independent over k , and suppose $P > N$. If the Zariski closure of the image of Φ has dimension at most N , then $I = \langle 1 \rangle$.*

Proof. Since $p_1, \dots, p_P$ satisfy no nontrivial polynomial relation over k , the point $\mathbf{p}$ is a generic point of $\mathbb{A}^P$. The attainable tool positions lie in a constructible set whose closure $\overline{\text{im } \Phi}$ has dimension at most $N < P$, hence is a proper closed subset of $\mathbb{A}^P$ and is contained in the zero set of some nonzero $F \in k[\mathbf{p}]$. Independence gives $F(\mathbf{p}) \neq 0$, so no $\mathbf{t}$ satisfies the generators and the ideal is the unit ideal. $\square$

Corollary 4. *A nonempty finite solution set requires $P = N$.*

Remark 5. The implementation uses Corollary 4 as a precondition test, rejecting $P \neq N$ by counting before any completion is attempted. Under this restriction the three-parameter case is the smallest one in which a manipulator positions a tool in space.

3 Experimental design

3.1 Substitution of the coefficient field

The discriminating measurement replaces $\mathbb{Q}$ by the prime field $\mathbb{F}_p$ with $p = 2^{31} - 1$, holding fixed the algorithm, the monomial order, the generators, the saturation and the number of adjoined parameters. In $\mathbb{F}_p$ an addition or multiplication is one machine-word operation followed by a reduction, and coefficients do not grow, so any cost surviving the substitution is independent of coefficient size. The substitution therefore separates the two accounts of Section 1 directly.

The field $\mathbb{F}_p(\mathbf{p})$ is implemented as a quotient of two polynomials over $\mathbb{F}_p$, normalised in the same three steps used for $\mathbb{Q}(\mathbf{p})$, which divide out the monomial content, divide out the polynomial gcd, and scale the denominator to be monic. The remainder of the library is templated on the coefficient type and is used without modification, so the two runs execute the same instruction sequence up to the arithmetic itself.

3.2 Validity control

Reduction modulo a prime preserves the structure of a Gröbner basis for all but finitely many primes. The exceptional primes are those for which the reduced basis acquires a different initial ideal, and detecting them is the standard obstacle in modular Gröbner methods [1]. We therefore record, for every completed run, the multiset of leading monomials of the reduced basis.

Remark 6. Agreement of the initial ideals between the two fields is a necessary condition for the substitution to be valid on a given input. Disagreement would indicate an unlucky prime and would invalidate a comparison of running times on that input, the two runs having completed different ideals.

3.3 Systems

Three systems are measured, each posed as in Definition 1 with one adjoined coordinate per joint, so that $P = N$ throughout in accordance with Corollary 4. The first is a planar two-link arm with two revolute joints about z and unit links, constrained in x and y , which has two solutions. The second is the shoulder and elbow of an anthropomorphic arm, obtained by removing its base joint in the manner of Pieper’s decoupling [18] and constrained by a radius and a height; it also has two solutions. The third is the anthropomorphic arm itself, a base yawing about z followed by two joints pitching about y , with unit links and all three coordinates constrained. Completing that system over $\mathbb{Q}$ at a fixed rational pose establishes independently that it has four solutions. The first two systems act as controls in the sense of Remark 6.

3.4 Instrumentation

Three quantities are recorded beyond wall-clock time. The completion is timed separately from the construction of the forward map and the residuals, since the latter also performs coefficient arithmetic and would otherwise be conflated with the former. The main loop of the completion algorithm reports the number of pairs processed, the current basis size, the pending pair count and the largest basis polynomial. The normalisation of $k(\mathbf{p})$ reports the largest numerator or denominator it has encountered, measured in terms.

3.5 Environment

Measurements were taken on an Intel Core i7-10870H at 2.20 GHz running Linux 6.8, compiled with g++ 11.4.0 at -O2, with GMP providing arbitrary-precision rational arithmetic. In the gcd measurements of Section 4.3, points whose operands have total degree at most eight are the median of five runs and larger points are single runs; the quantity of interest spans five orders of magnitude and is insensitive to run-to-run variation at that scale.

4 Results

4.1 Controls

Table 1 reports the two-parameter systems. Basis size, quotient dimension, largest basis polynomial and initial ideal agree exactly between the fields, so by Remark 6 the prime is not unlucky for these inputs. The substitution reduces the completion time by a factor of approximately four.

Table 1: Two-parameter systems, both fields. Times in seconds. *Setup* covers construction of the rational forward map and the residuals; *completion* covers saturation and the Gröbner basis. The initial ideal is given by the leading monomials of the reduced basis.

System	Field	Setup	Completion	Basis	$\dim_k A$	Max terms	Initial ideal
Planar 2R	$\mathbb{F}_p(x, y)$	0.000	0.009	2	2	3	$\langle t_2^2, t_1 \rangle$
Planar 2R	$\mathbb{Q}(x, y)$	0.001	0.031	2	2	3	$\langle t_2^2, t_1 \rangle$
Reduced 2R	$\mathbb{F}_p(r, z)$	0.000	0.008	2	2	3	$\langle t_2^2, t_1 \rangle$
Reduced 2R	$\mathbb{Q}(r, z)$	0.001	0.033	2	2	3	$\langle t_2^2, t_1 \rangle$

4.2 The three-parameter system

The anthropomorphic 3R system with all three coordinates constrained produced no result over $\mathbb{F}_p(x, y, z)$ within 400 s, nor over $\mathbb{Q}(x, y, z)$ within 900 s. Construction of the forward map and the residuals completed in under 0.05 s in both fields, so the whole of the elapsed time is attributable to the completion.

Instrumenting the main loop localises the failure. Over $\mathbb{F}_p(x, y, z)$ the run enters the loop with four basis elements, six pending pairs and a largest basis polynomial of fourteen terms, and processes one pair. No second pair was processed in the subsequent 300 s. The run is therefore inside a single reduction, and the failure is not an accumulation of many basis elements.

Instrumenting the normalisation identifies what grows during that reduction. The largest parameter polynomial appearing as a numerator or denominator reaches 460 terms after 93 s and 498 terms after 129 s, with intervals of tens of seconds between consecutive normalisations. Coefficients in $\mathbb{F}_p$ do not grow, so these intervals measure the cost of a gcd in $\mathbb{F}_p[x, y, z]$ on operands of that size. This observation motivates the measurement of the next section and fixes $m \approx 500$ as the operating point of interest.

4.3 The subresultant gcd in isolation

Table 2 and Figure 1 report the cost of a single gcd outside the pipeline. Two dense trivariate polynomials of equal total degree are formed with a planted common factor, so that the remainder sequence runs to full length rather than terminating early on coprime inputs. The same integer coefficients are used over both fields.

Table 2: One subresultant gcd on dense trivariate operands with a planted common factor. Here m is the number of terms per operand and t the running time in seconds.

Degree	m	gcd terms	t over $\mathbb{F}_p$	t over $\mathbb{Q}$	Ratio
4	35	10	0.000505	0.005692	11.3
6	84	20	0.008278	0.099993	12.1
8	165	35	0.114564	1.519231	13.3
10	286	56	1.073065	14.463953	13.5
12	455	84	6.554416	91.276337	13.9
14	680	120	31.064855	—	—
16	969	165	124.013371	—	—

Fitting $\log t = \alpha \log m + \beta$ by least squares over the points of Table 2 gives $\alpha_{\mathbb{F}_p} = 3.79$ and $\alpha_{\mathbb{Q}} = 3.81$. The ratio $t_{\mathbb{Q}}/t_{\mathbb{F}_p}$ rises from 11.3 at $m = 35$ and settles near 13.9 at $m = 455$.

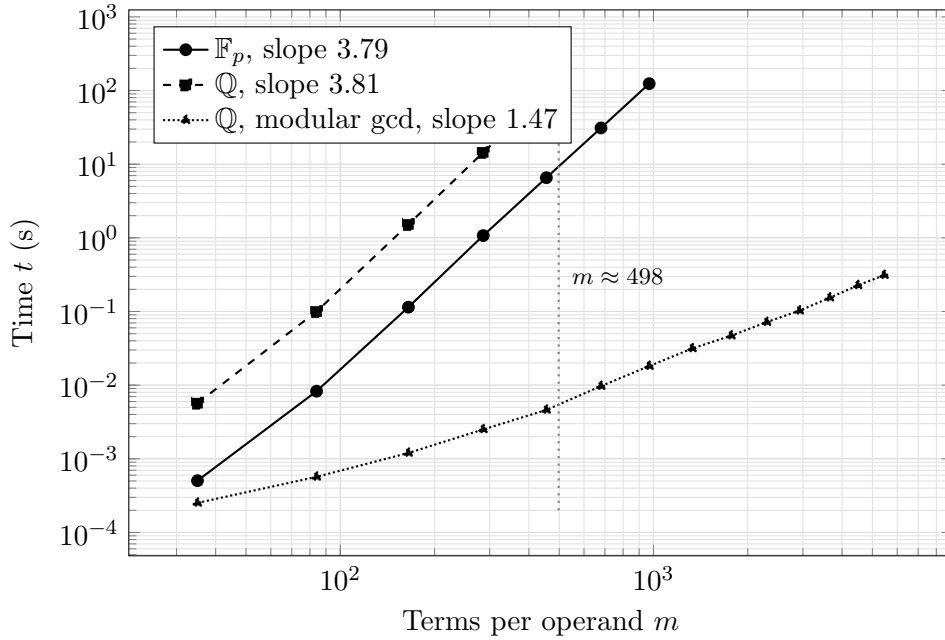


Figure 1: Cost of one gcd against operand size, both axes logarithmic. The two subresultant series are parallel, so the coefficient field contributes a multiplicative constant and leaves the exponent unchanged. The vertical line marks the operand size observed during the failing three-parameter run, which lies inside the measured range. The third series is the modular gcd of Section 6 over $\mathbb{Q}$ on the same operands (Section 9), whose slope is the one that changed.

5 Analysis

5.1 Separation of the two effects

The controls of Table 1 establish that the substitution of $\mathbb{F}_p$ for $\mathbb{Q}$ preserves the computation on the two-parameter inputs, by the criterion of Remark 6. The three-parameter system then fails to terminate in a setting where coefficient arithmetic is a machine-word operation and coefficients do not grow, which is inconsistent with the first of the two accounts as an explanation of the failure.

Table 2 quantifies what remains. The two fitted exponents agree to within 0.02, so the coefficient field affects the constant β and leaves α unchanged, which appears in Figure 1 as two parallel lines. The exponent $\alpha \approx 3.8$ is a property of the subresultant remainder sequence applied to multivariate operands. The sequence performs a number of pseudo-divisions that grows with the degree in the main variable, each pseudo-division is a polynomial multiplication and exact division over the remaining variables, and the content computations recurse into those variables in turn [4, 2]. The measured α is consistent with that structure and is present over both fields.

5.2 Consequences for modular gcd algorithms

The term *modular gcd* covers two constructions with different consequences here. The first computes the same subresultant remainder sequence over several primes and reconstructs the result by the Chinese remainder theorem, with a trial division to certify the candidate [2]. It addresses coefficient growth and leaves the sequence itself intact, so its benefit is bounded by the ratio measured in Table 2, approximately a factor of 13. Applied to a computation that has not completed in 900s, a factor of 13 does not produce a usable result.

The second reduces the multivariate problem to univariate gcds at sampled points and recovers the answer by interpolation, in the dense form due to Brown [2] or the sparse form due

to Zippel [24], with the Hensel-based variant of Moses and Yun [16] a further alternative. Such a construction replaces the remainder sequence and therefore acts on α rather than on β , which is what the measurement identifies as the governing quantity.

5.3 The operating point

Interpolating Table 2 between $m = 455$ and $m = 680$ at the exponent $\alpha = 3.79$ places a single gcd at $m = 498$ at approximately 9 s over $\mathbb{F}_p$ and, applying the measured ratio, approximately 125 s over $\mathbb{Q}$. Both figures agree with the intervals between consecutive normalisations reported in Section 4.2. Since $m = 498$ lies inside the range covered by Table 2, the estimate requires no extrapolation.

6 An interpolating gcd

The construction adopted is Brown's dense modular algorithm [2], in the form given in [9, Sec. 7.4], with two departures that the measurements of Section 9 show to matter. The certificate is an exact division over $\mathbb{Z}$ rather than over $\mathbb{Q}$, and the quotients of that division are returned to the caller. This section states the algorithm and the two facts about images on which its correctness rests.

Throughout, $A, B \in \mathbb{Q}[\mathbf{p}]$ are nonzero, $\prec$ is the monomial order under which the result is to be monic, and $G = \gcd(A, B)$ is normalised to be monic under $\prec$. Clearing denominators and dividing out the integer content writes $A = c_A A'$ with $c_A \in \mathbb{Q}$ and $A' \in \mathbb{Z}[\mathbf{p}]$ primitive, and likewise for B . By Gauss's lemma the primitive integer multiple G' of G divides A' and B' in $\mathbb{Z}[\mathbf{p}]$ [22, Sec. 6.2].

6.1 Images and their leading monomials

The algorithm maps the problem into simpler rings twice, first modulo a prime and then by specialising a parameter, and computes a gcd there. What makes the approach work is that an image gcd can be too large but never too small, and that being too large is visible in its leading monomial.

Proposition 7. *Let ℓ be a prime not dividing $\text{lc}_{\prec}(A')$, and let H_{ℓ} be the gcd of the images $\bar{A}', \bar{B}'$ in $\mathbb{F}_{\ell}[\mathbf{p}]$. Then $\text{LM}_{\prec}(H_{\ell}) \succeq \text{LM}_{\prec}(G)$, with equality if and only if H_{ℓ} is a nonzero scalar multiple of $\bar{G}'$.*

Proof. Since G' divides A' in $\mathbb{Z}[\mathbf{p}]$ and leading coefficients are multiplicative, $\text{lc}_{\prec}(G')$ divides $\text{lc}_{\prec}(A')$, so ℓ does not divide it and $\text{LM}_{\prec}(\bar{G}') = \text{LM}_{\prec}(G)$. Reduction is a ring homomorphism, so $\bar{G}'$ divides $\bar{A}'$ and $\bar{B}'$ and hence $H_{\ell} = \bar{G}'Q$ for some $Q \in \mathbb{F}_{\ell}[\mathbf{p}]$. Leading monomials are multiplicative over a field, so $\text{LM}_{\prec}(H_{\ell}) = \text{LM}_{\prec}(G) \text{LM}_{\prec}(Q) \succeq \text{LM}_{\prec}(G)$, with equality exactly when Q is a nonzero constant. $\square$

Proposition 8. *Let $A, B \in \mathbb{F}_{\ell}[p_1, \dots, p_k]$ be primitive when read as polynomials in $p_1, \dots, p_{k-1}$ over $\mathbb{F}_{\ell}[p_k]$, let lc denote the leading coefficient in that reading under the lexicographic order, and let $\gamma = \gcd(\text{lc } A, \text{lc } B) \in \mathbb{F}_{\ell}[p_k]$. For $\alpha \in \mathbb{F}_{\ell}$ with $\gamma(\alpha) \neq 0$, the gcd of $A|_{p_k=\alpha}$ and $B|_{p_k=\alpha}$ has leading monomial at least that of $G|_{p_k=\alpha}$, where $G = \gcd(A, B)$, with equality if and only if it is a scalar multiple of $G|_{p_k=\alpha}$.*

Proof. $\text{lc } G$ divides both $\text{lc } A$ and $\text{lc } B$, hence divides γ , so it does not vanish at α and specialisation preserves the leading monomial of G . The remainder of the argument is that of Proposition 7 with specialisation in place of reduction. $\square$

A prime or point whose image gcd has a leading monomial strictly larger than the true one is called *unlucky*. There are finitely many of each, since an unlucky prime divides a nonzero integer determined by A and B and an unlucky point is a root of a nonzero polynomial in p_k , in both cases a subresultant [2]. Propositions 7 and 8 turn this into an operational rule, under which the images with the smallest leading monomial among those computed are believed and any image with a larger one is discarded. An image with a smaller leading monomial than the current one proves every image held so far unlucky, and the accumulated work is restarted from it.

6.2 Recursion over a prime field

Over $\mathbb{F}_\ell$ the gcd of $A, B \in \mathbb{F}_\ell[p_1, \dots, p_k]$ is computed recursively in k . The contents of A and B as polynomials in $p_1, \dots, p_{k-1}$ over $\mathbb{F}_\ell[p_k]$ are univariate gcds and are removed first, their gcd being restored at the end. The remaining primitive parts are specialised at points α with $\gamma(\alpha) \neq 0$, the gcd of the specialisations is computed by the same procedure with $k - 1$ variables, and each such image, which is monic, is scaled so that its leading coefficient is $\gamma(\alpha)$. The scaling makes the images agree on a normalisation, namely $\gamma G / \text{lc}(G)$, whose degree in p_k is at most $\deg \gamma + \min(\deg_{p_k} A, \deg_{p_k} B)$; that many points plus one determine it, and it is recovered coefficient by coefficient by Newton interpolation. At $k = 1$ the problem is univariate and is a Euclidean algorithm on machine words.

The degree bound is a worst case, and the interpolant usually stops changing well before it is reached. The implementation therefore tests a candidate as soon as a new point leaves the interpolant unchanged, as well as once the bound is passed, and accepts the primitive part of the interpolant only if it divides both operands in $\mathbb{F}_\ell[p_1, \dots, p_k]$. A candidate that fails is not an error but a signal to take more points. Section 9 measures the effect of this early termination.

Polynomials over $\mathbb{F}_\ell$ are held as term lists sorted lexicographically with p_1 most significant. At level k the terms that agree in $p_1, \dots, p_{k-1}$ are then contiguous and ordered by their exponent of p_k , so that reading a polynomial as one with coefficients in $\mathbb{F}_\ell[p_k]$, and specialising p_k , are single passes over the list with no sorting.

6.3 Lifting to the rationals

Primes are taken in decreasing order below 2^{31} , so that a product of two residues fits a machine word, and a prime dividing $\text{lc}_{\prec}(A')$ or $\text{lc}_{\prec}(B')$ is skipped as Proposition 7 requires. Each image is made monic under $\prec$, which is the normalisation of G , so that the images of lucky primes are the reductions of G itself coefficient by coefficient. Images sharing the smallest leading monomial are combined by the Chinese remainder theorem, monomial by monomial over the union of their supports, and each combined coefficient is mapped back to a rational number by rational reconstruction [23, 22].

A reconstruction is only a candidate. It is tested once two consecutive reconstructions agree, since the test is the costly step and a reconstruction that is still changing is not worth testing, and the test is exact, requiring the primitive integer form G'' of the candidate to divide A' and B' in $\mathbb{Z}[\mathbf{p}]$. By Gauss's lemma divisibility over $\mathbb{Q}$ and over $\mathbb{Z}$ coincide for a primitive divisor, so a quotient coefficient that fails to be an integer already decides the question. The division keeps its remainder in an ordered map and subtracts each multiple of G'' in place, so that a step costs the number of terms of the divisor rather than of the remainder, and it stops at the first term that does not divide.

Proposition 9. *The procedure returns $\text{gcd}(A, B)$, monic under $\prec$, and terminates.*

Proof. Every image combined into a candidate C has the same leading monomial L and coefficient one there, and none has a term above L , so $\text{LM}(C) = L$, and $L \succeq \text{LM}(G)$ by Proposition 7. If C is returned it divides A and B , hence divides G , so $L \preceq \text{LM}(G)$. Therefore $\text{LM}(C) = \text{LM}(G)$, the quotient G/C is a constant, and it is one because both are monic. For termination, all

but finitely many primes are lucky, the images of lucky primes are the reductions of G , and rational reconstruction recovers every coefficient of G once the product of the primes exceeds twice the square of the largest numerator or denominator among them [22, Sec. 5.10]. From then on consecutive reconstructions agree and equal G , and the test succeeds. $\square$

Two consequences are used by the caller. If the image modulo any single admissible prime is constant, Proposition 7 shows that G is constant, and the procedure returns after one prime without a trial division; most calls made by normalisation end this way. And the division that certifies G has A'/G'' and B'/G'' as its quotients, from which the cofactors A/G and B/G follow by a rational scaling. Normalisation needs exactly these cofactors, so the procedure returns them with G , and no second division is performed.

The implementation formerly placed a heuristic in front of the gcd, which specialised every parameter but one at fixed values, examined the resulting univariate gcds for a common factor, and skipped the exact gcd when none appeared. That test gave evidence of coprimality rather than proof, which was admissible only because a skipped cancellation costs size and never correctness. The first consequence above supplies proof at a comparable price, and the heuristic is removed, so that every coefficient is now held in lowest terms.

7 Reconstruction from fixed poses

With the gcd replaced the three-parameter problem finishes on the anthropomorphic arm, and Section 9 shows that on arms with offsets it still does not. The parameter polynomials carried through the completion then reach 16 000 terms, the gcd accounts for a quarter of the time rather than most of it, and the rest is arithmetic on those polynomials. The action matrices the computation would produce on that arm have entries of at most fourteen terms. The intermediate expressions are the cost, and none of them is needed for the answer.

The action matrices $M_1, \dots, M_N$ and the normal forms of the variables, whose entries are the rational functions of the pose that the emitted solver evaluates, are therefore recovered without a Gröbner basis over $\mathbb{Q}(\mathbf{p})$. The approach is black-box rational function reconstruction [11], of the kind that has made finite-field methods standard in the reduction of Feynman integrals [21, 17].

7.1 The black box

Fix a prime ℓ and a pose $\pi \in \mathbb{F}_\ell^P$. The kinematic ideal at π is generated by $\Phi_{c_j}(\mathbf{t}) - D(\mathbf{t}) \pi_j$ and saturated by $\prod_i (1 + t_i^2)$ exactly as in Definition 1, but its coefficients are elements of $\mathbb{F}_\ell$ and it has no parameters. Its reduced Gröbner basis, standard monomials, action matrices and normal forms are computed by the same generic code instantiated over $\mathbb{F}_\ell$, and returned as a vector of $Nd^2 + Nd$ field elements, where $d = \dim A$.

The standard monomials at a general pose are determined once, from three rational poses solved over $\mathbb{Q}$, as the set on which two of them agree. A sample whose standard monomials differ is on a special fibre and is discarded. For π outside a proper Zariski-closed set, the reduced basis at π is the specialisation of the reduced basis over $\mathbb{Q}(\mathbf{p})$, and so are the matrices, entry by entry; this is the property that makes the samples values of the functions sought.

7.2 Reconstruction modulo a prime

Each entry is a rational function $f = n/m$ of the pose, in lowest terms. Its degrees are found first, on a random line $\pi(s) = u + sw$, along which f is a univariate rational function whose numerator and denominator degrees are, for a line in general position, the total degrees of n and m . They are read off by interpolating the samples along the line and running the extended Euclidean algorithm against the product of the $(s - s_i)$, choosing the step with the quotient

of largest degree [15], and the number of samples on the line is increased until every entry is reconstructed with at least two samples to spare.

Proposition 10. *Let $f = n/m$ in lowest terms with $\deg n = a$ and $\deg m = b$. The pairs (n', m') of polynomials with $\deg n' \leq a$, $\deg m' \leq b$ and $n'm = nm'$ form a one-dimensional space.*

Proof. From $n'm = nm'$ and $\gcd(n, m) = 1$, m divides m' ; by degree $m' = cm$ for a constant c , and then $n' = cn$. $\square$

With the degrees known, each sample (π_r, v_r) gives one linear equation

$$\sum_{|\alpha| \leq a} n'_\alpha \pi_r^\alpha - v_r \sum_{|\beta| \leq b} m'_\beta \pi_r^\beta = 0$$

in the coefficients of n' and m' over all monomials of total degree at most a and b . By Proposition 10 the solution space of the full system is a line, and a system built from enough random samples has the same solution space except on a proper algebraic subset of the choices of samples. The kernel is found by Gaussian elimination over $\mathbb{F}_\ell$ and normalised so that the coefficient of the leading monomial of m' under the graded reverse lexicographic order is one. That is how f is normalised over $\mathbb{Q}$, so the normalised vector modulo every prime that preserves this leading monomial is the reduction of the same rational vector.

The samples on the line are not used in these systems, since points on one line do not determine a polynomial in several variables, and a system filled with them has a kernel larger than a line. After the first prime the support of each entry is known, and later primes solve only for the monomials the first prime found, which on a sparse answer shrinks both the systems and the number of samples they need. A term that the first prime missed by vanishing modulo it makes the smaller system inconsistent, and the dense systems are then used again.

7.3 Lifting and checking

The normalised coefficients modulo successive primes are combined by the Chinese remainder theorem and mapped to rationals by rational reconstruction, as in Section 6, until two consecutive primes leave every reconstruction unchanged. Unlike the modular gcd, the result cannot be certified by a division, because the entries exist only as values. It is instead checked exactly. At two of the rational poses already solved over $\mathbb{Q}$, or more if the caller asks, the reconstructed entries are evaluated and compared with the entries of the matrices computed there, and any disagreement, or a pole of the reconstruction where $\mathbb{Q}$ has a value, is reported as a failure rather than returned.

Remark 11. The check poses have coordinates drawn from the $2^{21} + 1$ dyadic rationals $k/1024$ with $|k| \leq 2^{20}$. A wrong entry $\tilde{f} = \tilde{n}/\tilde{m} \neq f = n/m$, reconstructed under the same degree bounds (a, b) , agrees with f at a pose only if the pose is a zero of $\tilde{n}m - n\tilde{m}$, a nonzero polynomial of total degree at most $a + b$, which is 16 on the arms of Section 9. If the check poses are regarded as drawn uniformly and independently of the reconstruction, the Schwartz–Zippel lemma [20, 24] bounds the probability of that by $16/(2^{21} + 1) < 7.7 \times 10^{-6}$ per pose, and the probability that an error survives two poses by 6×10^{-11} . The implementation fixes the seed, so that a generated header does not depend on when it was generated; the bound describes the choice of seed, and the check does not replace a proof (Section 10).

8 Refining the emitted solver

The emitted solver forms a separating combination $\sum_i c_i M_i$ of the action matrices at the requested pose, computes the eigenvectors of its transpose, which are the evaluation functionals at the solutions, and reads each solution off an eigenvector through the normal forms of the variables.

The eigenvectors are computed to backward stability only. Near a pose at which two solutions give the separating form nearly the same value, the eigenvectors, and the solutions read from them, lose accuracy that the matrices themselves did not, and an emitted solver whose matrices are exact to rounding returns points with residuals many orders of magnitude above it.

Both routes have at hand the equations as posed, the polynomials $\Phi_{c_k}(\mathbf{t}) - D(\mathbf{t})p_k$ in $\mathbf{t}$ over $\mathbb{Q}$, in which the pose enters linearly. They are emitted into the solver with their Jacobian, differentiated exactly, and each solution returned by the eigenvalue method is refined by at most two Newton steps, a step being kept only if it lowers the norm of the residual. A solution already at rounding level is returned unchanged, and near a singular configuration, where the Jacobian is unreliable, a step that would increase the residual is not taken. The refinement changes nothing about the count of solutions, which remains $\dim A$.

9 Evaluation

The environment is that of Section 3. The programs are listed in Appendix A.

9.1 The gcd in isolation

Table 3 repeats the measurement of Table 2 with the modular gcd of Section 6, on the same operands over $\mathbb{Q}$, and carries it to operands eleven times as large. Figure 1 shows it as a third series.

Table 3: One gcd over $\mathbb{Q}$ on the dense trivariate operands of Table 2, computed by the subresultant sequence (taken from Table 2) and by the modular gcd. Times in seconds.

Degree	m	Subresultant	Modular	Ratio
4	35	0.005692	0.000252	23
6	84	0.099993	0.000569	176
8	165	1.519231	0.001201	1265
10	286	14.463953	0.002504	5776
12	455	91.276337	0.004613	19 787
14	680	–	0.009704	–
16	969	–	0.018164	–
18	1330	–	0.031478	–
20	1771	–	0.046988	–
22	2300	–	0.071854	–
24	2925	–	0.102573	–
26	3654	–	0.153842	–
28	4495	–	0.226325	–
30	5456	–	0.311915	–

Fitting $\log t = \alpha \log m + \beta$ over all points gives $\alpha = 1.47$ for the modular gcd, against 3.81 for the subresultant sequence over the same field, and the ratio between the two reaches about 19800 at $m = 455$, the largest size at which both were measured. At the operating point $m \approx 500$ of Section 4.2, a gcd that was estimated at 125 s takes about 5 ms.

9.2 The three-parameter solve

With the modular gcd in place, Table 4 repeats the experiment of Table 1 over $\mathbb{Q}$, and adds the three-parameter system, which now completes. Its basis has six elements, $\dim A = 4$, and its initial ideal is the one the fixed-pose solve reports, so the four branches known for the anthropomorphic arm are all accounted for.

Table 4: The systems of Table 1 over $\mathbb{Q}(\mathbf{p})$ with the modular gcd, and the three-parameter system. Completion time in seconds; the initial ideal of the last row is $\langle t_3^2, t_2 t_3, t_2^2, t_1 t_3, t_1 t_2, t_1^2 \rangle$.

System	Field	Completion	Basis	dim A	Max terms
Planar 2R	$\mathbb{Q}(x, y)$	0.014	2	2	3
Reduced 2R	$\mathbb{Q}(r, z)$	0.015	2	2	3
Anthropomorphic 3R	$\mathbb{Q}(x, y, z)$	19.36	6	4	5

The gcd is still a large part of the three-parameter solve, but no longer nearly all of it. It takes 9.4s of the 19.4s, over 7418 calls whose operands reach 1974 terms. Of those calls 2619 end in a trial division, none of which fails, and the rest are settled by a constant image modulo the first prime. The trial division itself takes 0.86 seconds. Table 5 measures the choices of Section 6 by removing them one at a time.

Table 5: The three-parameter solve of the anthropomorphic arm over $\mathbb{Q}(x, y, z)$ with parts of Section 6 removed one at a time. *Degree bound only* omits the early termination and the divisibility test inside the recursion, *trial division over $\mathbb{Q}$* performs the certifying division with the general division routine over $\mathbb{Q}$ and takes its quotients as the cofactors, and *coprimality heuristic* restores the specialisation test formerly placed in front of the gcd. Times in seconds.

Configuration	Completion	In gcd	Trial division	Evaluation points
As described	19.4	9.4	0.86	344 499
Degree bound only	22.4	12.8	0.82	859 985
Trial division over $\mathbb{Q}$	29.3	18.1	8.95	344 499
Coprimality heuristic restored	21.1	9.7	0.91	336 403
Subresultant gcd (Section 4)	> 900			

Early termination with a divisibility test halves the number of evaluation points, and the test never rejects on this input, which is the point of trying the candidate as soon as the interpolant stops moving. The trial division over $\mathbb{Q}$ through the general division routine, which rewrites the whole remainder at every step, is ten times as costly as the same division over $\mathbb{Z}$ with the remainder updated in place. The coprimality heuristic, which skipped the exact gcd on evidence of coprimality, is slightly slower than no heuristic at all, because a constant image modulo one prime answers the same question with a proof at about the same cost.

9.3 Arms

Table 6 extends the measurement to five three-joint arms. All have a base yawing about z except the last. The first is the anthropomorphic arm; the second displaces the shoulder from the base axis as the demonstration arm of the library does; the third displaces the elbow a quarter of a link sideways, out of the plane of the arm; the fourth displaces the base off its own axis, so that the placement no longer commutes with the rotation; and the fifth has axes z , $(3/5, 4/5, 0)$ and $(2/7, 3/7, 6/7)$ with offsets in general position. The decoupling refuses the last three, since the displaced elbow and the general axes take the tool out of any plane through the base axis and the displaced base carries the axis itself around.

Where both finish, the symbolic and the reconstructed matrices agree in every entry. The symbolic solve does not finish on the three arms with offsets, and the diagnostic run on the arm with the displaced elbow shows why. After 840s the gcd had taken 203s, a quarter of the time, and its operands had reached 16458 terms, whereas the largest entry of the reconstructed matrices has 14. The reconstruction spends most of its time in the black box, one fixed-pose completion over $\mathbb{F}_\ell$ per sample; on the general arm, the slowest, that is 28.6s of 47.0 seconds, the linear systems 4.6s, and the exact checks over $\mathbb{Q}$ most of the remainder.

Table 6: Three-joint arms, all with $\dim A = 4$. *Decoupled* is the two-parameter solve after the base is swept out, where the arm admits it; *symbolic* the solve over $\mathbb{Q}(x, y, z)$ with the modular gcd, stopped after one hour where marked, and, marked $\dagger$, terminated by the system after 54 minutes with 29 GB in use; *reconstructed* the method of Section 7, with the number of primes, samples and the largest numerator and denominator degree. Every reconstruction passed its exact checks at two rational poses and, in addition, the eigenvector identity at three rational configurations. Times in seconds.

Arm	Decoupled	Symbolic	Reconstructed	Primes	Samples	Degrees
Anthropomorphic	0.015	19.67	0.24	2	110	4/4
Shoulder displaced	0.059	> 3600	2.47	3	344	6/6
Elbow displaced	–	> 3200 $\dagger$	0.55	2	114	4/4
Base off its axis	–	34.94	0.25	2	118	4/4
General	–	> 3600	47.03	6	1664	8/8

9.4 The emitted solvers

Table 7 measures what a caller pays on the demonstration arm with the displaced shoulder. It compares the decoupled solver, which the library generates during its build, with the full solver over $\mathbb{Q}(x, y, z)$, which only the reconstruction can produce, each with and without the refinement of Section 8. Targets are drawn by mapping random configurations forward, so that all are reachable; residuals are the distance from the target of the forward map of each returned configuration. A damped least-squares iteration with a finite-difference Jacobian and a tolerance of 10^{-10} m is the baseline.

Table 7: Generated solvers on the demonstration arm, 20 000 targets for the full solver and 71 658 returned configurations for the decoupled one. Times per call.

Solver	Median	99th centile	Worst residual	Configurations
Decoupled, eigenvalues only	0.9 μ s	1.3 μ s	6.5×10^{-8} m	all
Decoupled, refined	1.5 μ s	1.9 μ s	1.6×10^{-15} m	all
Full, eigenvalues only	4.3 μ s	6.5 μ s	4.3×10^{-3} m	all
Full, refined	6.2 μ s	8.3 μ s	2.0×10^{-6} m	all
Damped least squares, one seed	20 μ s	350 μ s	10^{-10} m	one, 91% of seeds
Damped least squares, restarted	420 μ s			299 of 300 sets

The refinement removes the residual tail of both solvers at a cost of about 0.6 μ s for the decoupled one and 1.9 μ s for the full one. The one remaining outlier of the full solver, at 2.0×10^{-6} m, is a target within 2×10^{-4} m of the base axis, where a circle of configurations is about to appear; every other target of the 20 000 has all its configurations within 10^{-12} m, and the two solvers return the same configurations on 19 999 of them. The full solver is four times as costly per call as the decoupled one and inherits the pole of the half-angle substitution discussed in Section 10, so the decoupled solver is the one to use where the arm admits it. The damped least-squares iteration is thirteen times as costly as the refined decoupled solver for one configuration, and recovering the whole set by restarting it from random seeds took about ten seeds a target, missed a branch on one target of three hundred within sixty seeds, and cost nearly three hundred times as much.

10 Threats to validity

Non-termination is observed, not proved. The statements that a computation does not finish are bounded observations, of 400 s and 900 s for the subresultant runs of Section 4, and one

hour for the symbolic runs on arms with offsets in Section 9, except the one with the displaced elbow, which the system terminated after 54 minutes with 29 GB of memory in use, a first attempt having been killed earlier. The instrumentation that shows where those runs spend their time, a single reduction in the first case and operands growing past 16 000 terms in the second, is the stronger evidence.

One prime for the substitution. The control of Remark 6 verifies the prime used for the field substitution only on the two-parameter inputs; on the three-parameter input no verification is available. An unlucky prime would be expected to alter the initial ideal and the course of the computation, not to turn a terminating computation into a non-terminating one.

Dense operands. The operands of Table 2 are dense of their total degree, whereas the parameter polynomials of the pipeline need not be. Density makes the number of terms an unambiguous size measure, at the cost of favouring the dense interpolation adopted over the sparse method of [24] in any comparison between the two, which was not made.

The reconstruction is checked, not certified. The modular gcd ends in an exact division and is always correct; the reconstruction of Section 7 has no such test, and its exact checks at rational poses bound the probability of an undetected error (Remark 11) without excluding it. The symbolic solve therefore remains the default of the implementation, and the reconstruction an option. On the two three-joint arms and the two two-joint problems where both finish, the two agree in every entry. A certificate, such as an a priori degree bound that would turn finitely many exact checks into a proof, is not provided.

Scope of the arms. Five three-joint arms were measured. They cover the shapes that decide the cost in this setting, links with and without offsets, a base that does and does not admit the decoupling, and axes in general position, but they are not a sample of industrial manipulators, whose positioning subsystems are mostly of the first two kinds. Arms with prismatic joints were not measured.

One completion algorithm. All Gröbner bases were computed with one implementation of Buchberger’s algorithm with the normal selection strategy under the graded reverse lexicographic order. A completion of the F4 family [8] would change the number and shape of the coefficient operations in the symbolic solve, and the cost of each sample in the reconstruction, where most of its time is spent; neither was measured.

The half-angle chart. The emitted full solvers inherit a pole of the half-angle substitution. A target reachable only with a joint at exactly π in one family of solutions has a quotient of smaller dimension, and on the arms with a base yawing about z such targets fill the whole plane $y = 0$, where the solver reports the pose as outside its chart. The decoupled solvers recover the base angle by an arctangent and are not affected.

One machine. All timings were taken on one machine, with the compiler and flags of Section 3. The ratios between methods, which are what the conclusions rest on, span one to four orders of magnitude and are not sensitive to the machine at that scale.

11 Conclusions

The three-parameter parametric position problem did not finish over $\mathbb{F}_p(x, y, z)$, so its cost was not in the arithmetic on rational coefficients. The subresultant gcd costs about the 3.8th power

of the operand size over both fields, the field contributing a factor of about 13, so that a modular gcd formed by running the same remainder sequence over several primes could not have helped, and a gcd built on evaluation and interpolation could. Brown’s dense algorithm, certified by exact division over $\mathbb{Z}$ and returning its cofactors, brings the exponent to 1.5 and the three-parameter solve of the anthropomorphic arm from no result in 900s to about 20s.

That exposes the next cost rather than ending the problem. On arms with offsets the symbolic solve still does not finish in an hour, and the reason is no longer the gcd but the size of the intermediate rational functions, which exceed the final ones by three orders of magnitude. Reconstructing the final ones from fixed-pose solves over prime fields avoids them entirely, and every three-joint arm measured is then solved in under a minute, most in under a second, including three the decoupling cannot treat, one with its axes in general position, and every result agrees exactly with an independent computation over $\mathbb{Q}$ at rational poses and configurations.

The emitted solvers are fast and, with a Newton step on the equations as posed, accurate to rounding. A decoupled solve costs $1.5\mu\text{s}$ and its worst residual over 71 658 configurations is $1.6 \times 10^{-15}\text{ m}$, while a full three-joint solve costs $6\mu\text{s}$. Where an arm admits the decoupling it remains the better route, exact, fastest to produce and to call, and free of the half-angle pole; where it does not, the full solve now exists. Certifying the reconstruction, a second chart for the half-angle substitution, and a faster fixed-pose completion over prime fields are the work that remains.

A Reproduction

From the repository root, with the workspace built by `colcon build`:

```
CXX="g++ -std=c++17 -O2"
INC="-Ivarietas_core/include -Ivarietas_codegen/include \
-Ivarietas_kinematics/include -Ivarietas_ik/include \
-Idoc/experiments -I/usr/include/eigen3"

$CXX doc/experiments/field_cost.cpp -o field_cost $INC -lgmpxx -lgmp
./field_cost planar      # two-parameter control, both fields
./field_cost reduced    # two-parameter control, both fields
./field_cost spatial-fp  # three parameters over  $F_p(x,y,z)$ ; does not terminate
./field_cost spatial-q   # three parameters over  $Q(x,y,z)$ ; Table 4

$CXX doc/experiments/gcd_cost.cpp -o gcd_cost $INC -lgmpxx -lgmp
./gcd_cost subresultant  # Table 2 and the fitted exponents
./gcd_cost interpolating # the modular gcd on the same operands

$CXX doc/experiments/spatial_cost.cpp -o spatial_cost $INC -lgmpxx -lgmp
./spatial_cost          # Table 5, reconstructed, with the exact checks
./spatial_cost --symbolic anthropomorphic # and the symbolic solve beside it

$CXX doc/experiments/solver_cost.cpp -o solver_cost $INC \
-Ibuild/varietas_demo/generated -lgmpxx -lgmp
./solver_cost           # the decoupled solver at run time, and the baseline

ros2 run varietas_urdf urdf_codegen \
  varietas_demo/urdf/anthropomorphic_offset_3r.urdf full_offset_3r.hpp \
  --name full_offset_3r_ik --reconstruct
$CXX doc/experiments/full_solver_cost.cpp -o full_solver_cost \
```

```

-I. -Ibuild/varietas_demo/generated -I/usr/include/eigen3
./full_solver_cost          # the full and decoupled solvers side by side

```

Adding `-DVARIETAS_EXPERIMENT_TRACE` to the first build reports the largest parameter polynomial encountered during normalisation over $\mathbb{F}_p(\mathbf{p})$, the source of the figure $m = 498$. The pair-queue statistics of Section 4.2 were obtained with a temporarily instrumented copy of the completion loop, and the ablations of Table 5 with copies of the headers of the modular gcd and of the rational function field, modified as the table describes. The modular gcd keeps its own count of calls, primes and time, which `field_cost` prints.

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